

Requirement	Classification	Key Deciding Factor
1. Patient Web Portal (Public HTTPS Traffic)	Application Gateway	Requires URL-based routing, SSL termination, WAF protection, and Layer 7 inspection.
2. Diagnostic Image Streaming (DICOM)	Load Balancer	Uses raw TCP traffic, requires session persistence, long-lived connections, and no content inspection (Layer 4).
3. Clinical API (Internal Hospital Traffic)	Load Balancer	Internal-only traffic requiring TCP port-based distribution, sub-millisecond performance, and no HTTP awareness (Layer 4).
4. Authentication Service (Multi-Tenant Header Routing)	Application Gateway	Requires header-based routing using x-Trust-ID, multiple backend pools, end-to-end TLS, and Layer 7 processing.
5. Legacy Lab Results SOAP Service	Load Balancer	Requires only failover/redundancy for a legacy TCP service, with no payload inspection, no SSL offload, and Layer 4 routing.
6. Admin Dashboard (Host-Based Routing + Autoscale)	Application Gateway	Requires host-based routing (admin.healthbridge.nhs.uk vs www.healthbridge.nhs.uk) on a single public IP, HTTPS support, and Layer 7 routing.

Key Decision Rule Used -

Load Balancer - TCP/UDP traffic only, failover, session persistence, high-performance Layer 4 distribution.

(Looks only at IP addresses and ports. It does **not** inspect URLs, hostnames, headers, or HTTP content.)

Application Gateway - URL-based routing, host-based routing, header inspection, SSL termination, WAF, Layer 7 intelligence.

(Understands HTTP/HTTPS traffic and can inspect URLs, hostnames, headers, cookies, etc.)